calber — Memory (RAM) Settings Sheet

7 November 2026 · read from the machine (`dmidecode`, kernel log) · goal: the sticks' rated 3200 MT/s, stable 24/7

THIS SYSTEM

Board	Gigabyte Z370 AORUS Gaming WIFI (DMI: "WIFI-CF") · BIOS **F15a** (11/28/2019) · PCB rev _____
CPU	Intel Core i7-9700F, 8 cores, no integrated graphics · Intel's rated memory speed: **DDR4-2666**
RAM	4 × 16 GB Team Group DDR4-3200 (`TEAMGROUP-UD4-3200`), **single rank**, all four slots (A-DIMM0/1, B-DIMM0/1)
Running now	**3200 MT/s since 7 Nov, 11:00 (was 2400 at 1.2 V, XMP off) · XMP Profile 1, DRAM 1.35 V**
Cooling	CPU on liquid: no air across the DIMMs or the VRM. Allow time for heat soak before judging temperatures.
XMP profile	Profile 1: speed **3200** · voltage **1.35 V** · timings as the profile sets them _____

SETTINGS — BIOS, TWEAKER TAB

Setting	Set to	24/7 limit
Extreme Memory Profile (X.M.P.)	**Profile1**	—
DRAM Voltage	**The profile's, usually 1.35 V**	≤ 1.40 V (stay at 1.35 V if stable)
Memory Enhancement Settings	Normal	—
Command Rate (tCMD)	Auto; **2T** if unstable	—
VCCSA (System Agent)	Auto; **1.15 V** if unstable, +0.05 V steps	≤ 1.25 V
VCCIO	Auto; **1.10 V** if unstable, +0.05 V steps	≤ 1.20 V
System Memory Multiplier	From the profile (32.00 = 3200)	Step down: 30.00, then 29.33

STEPS

1. □ Orrery → Projects → **Finish, then out**; wait for "Removed".
2. □ Reboot, **DEL**. Make the other changes first (CSM off, AC BACK, logo off, PXE off).
3. □ **F11** → **Save Profile "stock"** (the way back).
4. □ X.M.P. → Profile1. Fill in the profile line above.
5. □ DRAM Voltage = profile (1.35 V). Enhancement: Normal.
6. □ **F10**. First boot is slow; the board may cycle a few times.
7. □ Cycles 3× and comes back at defaults? **F11** → **Load "stock"**, go to *If unstable*.
8. □ Ubuntu: `sudo dmidecode -t 17 | grep Configured` should read **3200 MT/s, 1.35 V** on all four.
9. □ Stress test (calber still out of the queue), below.
10. □ Tell Claude → speed bench `ram-bench.sh calber 3200`.
11. □ **F11** → **Save Profile** + USB stick. Then Orrery **Add**.

IF UNSTABLE — ONE CHANGE AT A TIME

1. □ Command Rate **2T**.
2. □ VCCSA **1.15 V**, VCCIO **1.10 V**; then +0.05 V (limits above).
3. □ Speed **3000**, profile timings and voltage kept.
4. □ Speed **2933**.
5. □ Give up safely: **XMP off, 2666 at 1.2 V** (multiplier 26.66): the CPU's own rating, no test needed.

STRESS TEST (1 HOUR)

```
sudo apt install stress-ng
stress-ng --vm 4 --vm-bytes 75% --vm-method all \
  --verify -t 1h
```

Any "fail" or a crash = unstable. Stricter: MemTest86 from USB overnight.

> **Bad memory fails silently:** wrong census results or a damaged island checkpoint, not a crash. Do not skip the test at 3000+.

SPEED — BEFORE AND AFTER (SAME FROZEN WORK, CALBER IDLE)

Measure	2400, 1.2 V (before)	After: 3200	Change
Bandwidth, all 8 cores (triad)	22,284 MB/s	27,686 MB/s	+24%
Bandwidth, one core (triad)	20,291 MB/s	24,436 MB/s	+20%
Latency (one load, 512 MiB)	69.6 ns	61.5 ns	12% quicker
Census, 16 seeds on 8 cores	72.0 s (3 runs)	71.2 s (3 runs)	1.1% faster, same games
Census, 2 seeds on 1 core	16.1 s (2 runs)	16.0 s (2 runs)	0.9% faster
Fleet jobs, s/seed vs fleet (a day)	0.82 (17 jobs)	after a day	

Expect a small gain: calber's census showed no slowdown at 2400. VCCSA/VCCIO limits are common Coffee Lake practice, not checked against Gigabyte's or Intel's documents. Full reasoning: `docs/calber-bios-recovery-lab-report.md` §V; checklist: `docs/calber-todo.md`.